

Que1 :- output:

```
Console × Eclipse IDE for C/C++ Developers 2026-09 1
Assignment2.exe [C/C++ Application] C:\Users\Administrator\eclip

Enter details for Product 1
Enter Product ID: 1002
Enter Name: widget-A
Enter Price: 250.00
Enter Quantity: 120

Enter details for Product 2
Enter Product ID: 1002
Enter Name: Bolt-B
Enter Price: 15.50
Enter Quantity: 8

Enter details for Product 3
Enter Product ID: 1003
Enter Name: Widget-C
Enter Price: 450.00
Enter Quantity: 160

Enter details for Product 4
Enter Product ID: 1004
Enter Name: Bolt-D
Enter Price: 99.99
Enter Quantity: 55
```

```
Enter details for Product 5
Enter Product ID: 1005
Enter Name: T-E
Enter Price: 550
Enter Quantity: 99

===== INVENTORY REPORT =====
ID      Name      Price      Qty      Total Value
-----
1002    widget-A      250.00     120     30000.00
1002    Bolt-B        15.50       8       124.00
1003    Widget-C     450.00     160     72000.00
1004    Bolt-D        99.99       55     5499.45
1005    T-E          550.00       99     54450.00

Highest Value Product : 72000.00

Enter low stock threshold:
```

Que2 :- output:

```
Console × Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) Assignment2.exe [C/C++ Application] C:\Users\Administrator

Reorder Cost (int quantity) : 1000
Reorder Cost (double quantity) : 1050
Reorder Cost (with tax) : 1180

Price after default 10% discount : 900
Price after 20% discount : 800
```

Que3 :- output:

```
Console x Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) Assignment2.exe [C/C++ Application] C:\Users\Admin
===== HOSPITAL PATIENT REGISTRY =====
Default Constructor called
Parameterized Constructor called
Parameterized Constructor with fee called

===== PATIENT DETAILS =====

Patient ID      : 0
Patient Name    : Unknown
Age             : 0
Registration Fee : Rs. 500

Patient ID      : 101
Patient Name    : Tanaya
Age             : 22
Registration Fee : Rs. 500

Patient ID      : 102
Patient Name    : Apurva
Age             : 25
Registration Fee : Rs. 1000
```

```
Console x Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) Assignment2.exe [C/C++ Application] C:\Users\Administrator\
Registration Fee : Rs. 1000

Enter number of patients: 2
Default Constructor called
Default Constructor called

Dynamic Patient Array Created.

Patient 1:

Patient ID      : 0
Patient Name    : Unknown
Age             : 0
Registration Fee : Rs. 500

Patient 2:

Patient ID      : 0
Patient Name    : Unknown
Age             : 0
Registration Fee : Rs. 500
Destructor called for Patient ID: 0
Destructor called for Patient ID: 0

Dynamic Patient Array Deleted.
Destructor called for Patient ID: 102
Destructor called for Patient ID: 101
Destructor called for Patient ID: 0
```

Que4:- Output:

```
Console × Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) Assignment2.exe [C/C++ Application] C:\Users\Administrator\
===== GAME ENTITIES =====
Name   : Aragorn
Health : 100
Level  : 10
Type   : Player

Name   : Orc
Health : 60
Level  : 5
Type   : Enemy

Name   : HealthPotion
Health : 0
Level  : 1
Type   : Item

Player Name : Aragorn
Player Health: 100
Player Level : 10
Player Type  : Player
```

Que5:-output:

```
Console × Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) Assignment2.exe [C/C++ Application] C:\Users\Administrator\
Physics Clamp Result : 100
GameMath Clamp Result : 99
Physics Lerp Result   : 15
GameMath Lerp Result  : 25
Using Physics Namespace Lerp : 60
```

Que6:-output:

```
Console × Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) Assignment2.exe [C/C++ Application] C:\Users\Administrator\
Enter number of rows: 3
Enter number of columns: 3
|
===== GAME MAP (3 x 3) =====
2  0  3
1  3  3
0  1  1

Legend: 0=Grass  1=Water  2=Mountain  3=Forest  4=Dungeon

Tile Count:
Grass   : 2
Water   : 3
Mountain : 1
Forest  : 3
Dungeon : 0
```